A MORPHISM OF AFFINE FORMAL ALGEBRAIC SPACES IS OF FINITE TYPE EXACTLY WHEN THE RING MAP IS TAUT AND TOPOLOGICALLY OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one identification of quotients; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Tautness of the ring map corresponds to representability of the induced morphism. Same chapter, Lemma lemma-representable-affine, invoked at proof lines 6352-6354 and again at proof lines 6368-6370. Supplies representability and the description of the fibre products used in both directions.
- Morphisms from a quasi-compact scheme into a filtered colimit factor through a stage. Same chapter, Lemma lemma-factor-through-thickening, invoked at proof lines 6364-6366. Reduces the verification of finite type to a single quotient.
- Surjectivity of a ring map can be tested modulo a locally nilpotent ideal. Algebra, Lemma algebra-lemma-surjective-mod-locally-nilpotent, invoked at proof lines 6385-6387. Gives surjectivity of the polynomial map after passing to a second quotient.

Source reference. Original lemma 07RD: sources/stacks-chow/tags.txt, source lines 9863–9863. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN5: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4239–4239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $\varphi : A \to B$ be a continuous map of weakly admissible topological rings over S. The following are equivalent*

(1) *$Spf(\varphi) : Y = Spf(B) \to Spf(A) = X$ is of finite type,*
(2) *φ is taut and B is topologically of finite type over A.*

Proof. We can use Lemma 0AN5 to relate tautness of φ to representability of $\mathrm{Spf}(\varphi)$. We will use this without further mention below. It follows that $X = \operatorname{colim} \mathrm{Spec}(A/I)$ and $Y = \operatorname{colim} \mathrm{Spec}(B/J(I))$ where $I \subset A$ runs over the weak ideals of definition of A and $J(I)$ is the closure of IB in B.

Assume (2). Choose a ring map $A[x_1, \ldots, x_r] \to B$ whose image is dense. Then $A[x_1, \ldots, x_r] \to B \to B/J(I)$ has dense image too which means that it is surjective. Therefore $B/J(I)$ is of finite type over A/I. Let $T \to X$ be a morphism with T a quasi-compact scheme. Then $T \to X$ factors through $\mathrm{Spec}(A/I)$ for some I (Lemma 0AIA). Then $T \times_X Y = T \times_{\mathrm{Spec}(A/I)} \mathrm{Spec}(B/J(I))$, see proof of Lemma 0AN5. Hence $T \times_Y X \to T$ is of finite type as the base change of the morphism $\mathrm{Spec}(B/J(I)) \to \mathrm{Spec}(A/I)$ which is of finite type. Thus (1) is true.

Assume (1). Pick any $I \subset A$ as above. Since $\mathrm{Spec}(A/I) \times_X Y = \mathrm{Spec}(B/J(I))$ we see that $A/I \to B/J(I)$ is of finite type. Choose $b_1, \ldots, b_r \in B$ mapping to generators of $B/J(I)$ over A/I. We claim that the image of the ring map $A[x_1, \ldots, x_r] \to B$ sending x_i to b_i is dense. To prove this, let $I' \subset I$ be a second weak ideal of definition. Then we have

$$B/(J(I') + IB) = B/J(I)$$

because $J(I)$ is the closure of IB and because $J(I')$ is open. Hence we may apply Algebra, Lemma 07RD to see that $(A/I')[x_1, \ldots, x_r] \to B/J(I')$ is surjective. Thus (2) is true, concluding the proof. □